Industry Brief

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EU MDR Made Simple: Top Tips for Medical Device Compliance Success

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Medical devices without the MDR/IVDR CE Mark won't be permitted to be sold in the EU, which means they won't be available for use in health care systems.[1]

Nearly everyone will interact with a medical device in some way during their lifetime. It is therefore crucial to ensure these devices are safe, effective, and perform as intended. Yet, given the industry's velocity, it can be challenging for medical device companies to consistently produce high-quality devices when they must adapt quickly to regulatory changes.

What is EU MDR?

Recognising the need to enforce heightened safety measures and risk management processes for medical devices marketed and used in Europe, the European Parliament and the Council of the European Union developed the Medical Device Regulation (MDR) and In Vitro Diagnostic Regulation (IVDR) in 2017. These regulations replace the previous Medical Device Directive (MDD) established in the early 1990s and repealed in May of 2021. Compared to the previously instituted directives, the EU MDR asks device manufacturers to place more emphasis on a life-cycle approach to safety and to support that approach with clinical data.

The intent of the updated regulations is to increase transparency in the approval and usage of medical devices while improving patient experiences and outcomes. Under the most recent updates to the regulations, medical devices and in vitro devices must receive a CE Marking (the symbol for signifying European conformity to European regulations established by the European Commission [EC], European Parliament [EP], and European Union [EU] member states) certificate by the established deadlines.

Certain "legacy devices" placed on the market after the MDR effectivity date can benefit from an extended transitional period. To learn if your company's devices qualify for exemption under the extension of the EU MDR compliance transitional period, see this July 2024 EC Q&A.

Glossary of EU MDR/IVDR Terms

- **CER (Clinical Evaluation Report):** A technical document compiling the clinical evidence and evaluation data collected from investigations or studies on substantially equivalent devices.
- **EUDAMED (European Database on Medical Devices):** The EU's official web-based database for storing and exchanging information related to all aspects of medical device development.
- **GUDID (Global Unique Device Identification Database):** Administered by the U.S. Food and Drug Administration (FDA), this catalog serves as a reference for every device with a unique device identifier (UDI).

- **IMDRF (International Medical Device Regulatory Forum):** Established in 2011, this forum of medical device regulators from around the world discusses future directions in the harmonisation of medical device regulation.
- **MDD (Medical Device Directive):** The former medical device regulation for CE approval, now rescinded.
- **NB (Notified Body):** An organisation designated by an EU member state to assess product conformity and grant CE approval for medical devices.
- **PMCF (Postmarket Clinical Follow-Up):** A continuous process updating the clinical evaluation of a medical device, addressed in the postmarket surveillance plan.
- **PMS (Postmarket Surveillance):** Process for ensuring a device remains safe once on the market.
- **PSUR (Periodic Safety Update Report):** Required by the MDR for all Class IIa devices and above, this periodic report provides a risk-benefit evaluation of a medical at defined time points post-authorisation.
- **SSCP (Summary of Safety and Clinical Performance):** Required for certain devices, considering diagnostic or therapeutic options addressed in the CER, this document provides access to current clinical data as well as safety and clinical performance information about a medical device.
- **UDI (Unique Device Identifier):** A series of numeric or alphanumeric characters created via a globally accepted device identification and coding standard that allows for the unambiguous identification of a specific device on the market.

Overview of Changes in the Updated EU MDR Guidelines

- More premarket regulatory obligations.
- Stricter risk management requirements.
- Establishment of and phased rollout plan for a European medical device database.
- New identification and traceability requirements for existing and new devices.
- Expansion of medical device classifications to new areas.
- More postmarket regulatory obligations.
- New Notified Body (NB) accreditation requirements.
- Changes to technical documentation requirements.
- New requirements for quality management system (QMS) compliance.
- Requirement to notify authorities of supply disruptions or product discontinuations.

Premarket Regulatory Obligations

The EU MDR's essential premarket conformity requirements, referred to in Annex I as the General Safety and Performance Requirements (GSPR), have increased. The GSPR breaks down into four main categories:

1. How devices should be designed.
2. Requirements for safety and risk management.
3. Requirements for materials and packaging.
4. Information the manufacturer must supply with the device.[2]

To fulfill GSPR requirements, manufacturers must plan and implement a systematic process to continuously generate, collect, and analyse clinical data pertaining to a device in order to verify and sustain the device's safety and performance.

Risk Management

Under EU MDR guidelines, risk management must be an ongoing iterative process throughout a device's life cycle, including systematic updates. The intent of the requirements spelled out in Annex I of the MDR is "the reduction of risks as far as possible without adversely affecting the benefit-risk ratio."[3] For EU MDR compliance, manufacturers are required to:

- Establish and document a risk management plan for each device.
- Identify and analyse known and foreseeable hazards associated with each device.
- Estimate and evaluate risks that could occur from intended use and potential misuse.
- Eliminate or control all risks.
- Evaluate the impact of information from the production phase and postmarket surveillance system on the estimates of their associated risk, as well as on the overall risk, benefit-risk ratio, and risk acceptability.
- Following the evaluation of the information's impact, amend control measures if necessary.

Manufacturers will need to enter reports of all incidents, injuries and deaths involving their devices into EUDAMED.

Global Database for Medical Devices: EUDAMED

Medical devices are constantly evolving in performance and sophistication, which can present challenges in medical device development and usage. The development of the European Database for Medical Devices (EUDAMED) is intended to help alleviate these challenges and foster more harmonisation with device quality and safety requirements.

Established by Regulation (EU) 2017/745 on medical devices and Regulation (EU) 2017/746 on in vitro diagnosis medical devices, the database's aim is to:

- Facilitate the harmonisation of quality and safety requirements for medical devices by functioning as a central repository for information on all medical device product life cycles.

- Help European authorities conduct pre- and postmarket surveillance on medical devices through information exchange.
- Enhance overall transparency through better access to information for both the public and health care professionals.

The information contained in EUDAMED includes:

- Data related to the registry of manufacturers, authorised representatives and devices.
- Data related to the certificates issued, modified, supplemented, suspended, withdrawn or refused according to established procedures.
- Data obtained in accordance with the vigilance procedure on incidents or near-incidents, which occur during the use of the medical device.
- The summary of safety and clinical performance (SSCP) of medical devices, which is entered into the database by the NB.

Mandatory use of EUDAMED was initially expected to be fully enforced when the MDR went into effect. However, recent amendments to the legislation have facilitated a gradual, phased rollout of the database until the last of its six interconnected modules are completed. Since the recent MDR amendment eliminates the requirement that EUDMED's use would have become mandatory once the entire database was completed and audited, the EC has published a new roadmap that outlines the phased approach to the mandatory use of specific EUDAMED modules. The roadmap indicates that some processes may become mandatory by the end of 2025.[4] The overall intent of this shift to the phased rollout is to accelerate the system's use, improve transparency, and enhance the monitoring of device availability.[5]

New Device Identification and Traceability Requirements

To facilitate urgent action during events such as recalls, devices must comply with the UDI system, effectively identifying them throughout their product life cycle. Manufacturers must assign and maintain unique UDIs for their devices, and these identifiers will be required on all device labels under the EU MDR guidelines outlined in Annex VI Part C.

Medical Device Classification

In the updated EU MDR compliance requirements, many medical devices are classified differently than they were under the MDD. Depending on the device's purpose, some will be bumped up to a higher risk class. These include:

- Devices that incorporate nanotechnology.
- Orthopaedic implants like spinal disk replacements.
- Surgical meshes.
- Life-saving therapeutic devices like automated external defibrillators.[6]

Additionally, the classification rules outlined in Annex VIII specify that software influencing device use will fall within the same class as the device whilst software that is independent of any other device shall be classified in its own right.

EU MDR also expands the definition of a medical device to include non-medical and cosmetic devices not previously regulated, such as aesthetic products, cleaning products for disinfecting medical devices, and contraception products.

Reusable Devices

A new Class I category (Class Ir) has been created for reusable devices like surgical instruments and endoscopes. These devices will not be granted extensions, and grandfathering in legacy devices is prohibited.

IVDR Classification

IVDR introduced a risk-based classification for devices, with classifications based on intended purpose and potential health risks:

MDR Class	Risk Profile
A	Low individual risk and low risk to public health.
B	Moderate individual risk and/or low risk to public health.
C	High individual risk and/or medium risk to public health.
D	High individual risk and high risk to public health.

Class A devices are self-certified by manufacturers unless sold sterile. Class B, C, and D devices require a conformity assessment by the NB.

Conclusions from the review of PMS data must be summarised in a report. This report needs to be reviewed by the NB and passed to an expert panel, which may provide an opinion with respect to the device.

Postmarket Regulatory Obligations

Regulatory requirements for postmarket surveillance will become more granular and stringent. Once a product is on the market, the manufacturer will be required to continue gathering clinical data and examining the product through postmarket surveillance (PMS). As stated in the Article 83 of the EU MDR, the PMS system is used to:

- Update the benefit-risk determination and improve risk management.
- Update the design and manufacturing information, i.e., the instructions for use in the product's labelling.
- Update the clinical evaluation.
- Update the summary of safety and clinical performance.

- Identify needs for preventive, corrective, or field safety corrective action.
- Identify options to improve the device's usability.
- Contribute to the postmarket surveillance of other devices.
- Detect and report trends.[7]

For high-risk devices where long-term safety and performance data is unknown or where EC certification is based on equivalence, postmarket clinical follow-up (PMCF) data is required.

Notified Body Accreditation

The designation process could take up to 12 months or more. It's important to take this into account when planning the certification of your products.

The EU MDR brings more rigorous designation requirements for NBs along with increased control and monitoring by the EC and the national competent authorities.[8] Under the regulation's more stringent criteria, NBs must be redesignated and cannot perform recertifications until they have met the new designation requirements. To ensure an NB meets the new expectations for regulatory compliance, medical device companies should ask the following questions:

- Is your NB fully designated as an MDR approver?
- Does the scope of your NB's designation cover all your products?
- Does your NB have the availability to evaluate and approve your device by the established deadline?
- Does your NB require additional data on your devices?
- Is your NB knowledgeable about the transitional provisions of the updated regulation?

Technical Documentation

Technical documentation requirements have changed substantially under EU MDR. For starters, the regulation no longer includes the design dossier that was required by the MDD. The dossier information and all other documentation now falls under the category of technical documentation. Some of the key technical documents include:

- Risk analysis.
- Clinical data analysis.
- General safety and performance requirements.
- Quality assurance standard operating procedures (SOPs).
- Labelling updates.
- Postmarket clinical follow-up (PMCF), which includes the clinical evaluation report (CER) and periodic safety update report (PSUR).

Manufacturers should perform a gap analysis to determine how the EU MDR's clinical investigation requirements will impact the existing clinical investigation for new devices.

Quality System Compliance

Just as the FDA's quality system regulation (21 CFR Part 820) is inching toward ISO 13485 harmonisation, the MDR's quality management system (QMS) requirements now also largely correspond to the globally recognised quality standard. However, compliance to ISO 13485 doesn't guarantee that a medical device will be fully compliant with EU MDR. It's important to understand the new expectations and identify any areas where your current QMS fails to meet the updated regulation's requirements.

The FDA's quality system regulation requires that all companies developing regulated products must implement a QMS to ensure that finished devices will be safe, effective and in full compliance with the U.S. Federal Food, Drug and Cosmetic (FD&C) Act. Similarly, as more robust regulatory requirements are enforced by the MDR, companies will need to revisit their core quality processes – including quality assurance, risk management, and postmarket expectations – to ensure they are in compliance with the new EU MDR guidelines.

Supply Disruption Notifications

The MDR/IVDR guidelines have been amended to require manufacturers to notify authorities when they anticipate supply disruptions or if they intend to discontinue a product. The details about and reasons for supply disruptions must be shared with the competent authority in the EU country where the manufacturers are based with six months' advance notice. Health care institutions, professionals, and economic operators affected by the supply issues must also be notified.[9]

Doctors in hospitals need to have the necessary medical devices at the ready to perform procedures. Unless a device is fully compliant with EU MDR by the established deadlines, it cannot be used where it is needed.

Preparing for EU MDR Compliance

The medical device industry is advancing faster than ever. To keep up with the rapid pace of change, manufacturers need to be agile, well-versed in regulatory changes, and ready to adopt modern digital technologies for managing quality and regulatory compliance processes.

If your organisation isn't yet familiar with the updated EU MDR guidelines, it's vital to be aware of the following factors:

- NBs will not rely on historic product performance or reviews.
- All your data must be kept current.
- You must have immediate access to up-to-date documentation from new clinical studies.
- The certification time frame is tight, which means there is no margin for error and there are no do-overs in your pursuit of regulatory compliance.

Simplify and Accelerate EU MDR Compliance With Digital Solutions

MasterControl Quality Excellence: The Ideal MDR Software for Compliance

Keeping up with the momentum of regulatory changes is vital to success, so medical device companies need a quality management system that's built for compliance. Modernising your approach to compliance with an advanced digital QMS ensures preparation for EU MDR. MasterControl Quality Excellence – the number one QMS platform in the life sciences – is ideally suited for compliance with evolving MDR requirements. It is specifically designed to help medical device and other companies in the life sciences break down silos, eliminate errors, create efficiencies, reduce delays, and provide regulators and customers with the assurance of consistently high-quality, safe, and effective products.

Contact a MasterControl specialist to learn how Quality Excellence can help you achieve EU MDR compliance faster.

About MasterControl

MasterControl Inc. is a leading provider of cloud-based quality and compliance software for life sciences and other regulated industries. Our mission is the same as that of our customers – to bring life-changing products to more people sooner. The MasterControl Platform helps organisations digitise, automate and connect quality and compliance processes across the regulated product development life cycle. Over 1,000 companies worldwide rely on MasterControl solutions to achieve new levels of operational excellence across product development, clinical trials, regulatory affairs, quality management, supply chain, manufacturing and postmarket surveillance. For more information, visit www.mastercontrol.com.

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